

Ethan Granat

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POLYMER MATERIALS ENGINEER with broad hands-on experience in polymer, composite, and tooling fabrication and characterization. Developer of materials, products, and processes with full end-to-end project ownership. Driven by a passion for creative problem solving, continuous learning, and process improvement.

EDUCATION

Western Washington University

Bellingham, WA

BS, Polymer Materials Engineering (ABET Accredited)

Sept 2021 - June 2025

Concentration in Polymer Circularity | Minor in Chemistry

Capstone:

Engineered a novel cure process for a FROMP-curing resin composite system.

Optimized a three-minute out-of-autoclave cure cycle utilizing sequential 2³-run DOE trials, achieving 40°C glass transition temperature and 90% degree of cure in minimal cure time.

Prepared and characterized 100+ polymer composite samples by DSC and DMA.

Waste to Products Researcher:

Conducted literature review and feasibility study, in partnership with commercial venture, for batch pyrolysis replication of polymer film.

Processed and recycled 20kg PLA and HDPE feedstock per quarterly workshop of 100+ participants. Demonstrated lab processes and equipment to groups of 20 elementary school students in Compass2Campus higher education outreach program.

WORK HISTORY

Independent Contractor

Seattle, WA

Robotics Engineer

Feb 2026 – April 2026

- Integrated ROS2 software stack for a custom differential drive robot, combining SLAM Toolbox, Nav2, and ros2_control to achieve repeatable and reproducible floor mapping and autonomous indoor navigation.
- Configured web-based UI control and enabled waypoint following from desktop and mobile devices over Wi-Fi.

Zero Waste Western

Bellingham, WA

Event Coordinator

Sept 2022 – May 2025

- Hand re-sorted 50+lbs of campus waste stream samples, quantifying recycling and contamination rates across each of 3 core campus sectors.
- Translated data findings into targeted, data-driven waste sorting signage, educational campaigns, and workshops.
- Designed, tested, and led 25 zero waste workshops over 3 years, positively engaging over 500 students in waste reduction.
- Spearheaded social media and publicity campaigns (including a clothing repair fair by coordinating 10 volunteers), increasing Instagram engagement by 667% in one quarter and diverting 100+ garments from landfill.

Freelance & Additional Experience

Multiple Locations

Uncut Media	Set Design Assistant	New York, NY	Sept 2026
Lucho Effects	Production Assistant	Seattle, WA	Aug 2026
Shifting Locus	Production Assistant	Seattle, WA	May 2026
DoorDash	Delivery Driver	Bellingham, WA	Aug 2025 – Current
Shifting Locus	Extra	Seattle, WA	Jan 2024

TECHNICAL SKILLSETS

CAD, CAM, & CNC:

SolidWorks solid and surface modeling; HSMWorks 2.5-axis toolpath programming; VERICUT post-processing; Haas VF2 operation; FEA; design for manufacturing; injection mold fabrication; ROS2; Python.

Characterization:

Colorimetry; FTIR, NMR, and UV-Vis spectroscopies; thermomechanical properties, including tensile testing, melt-flow, TGA, DSC, DMA, and SEM (proficient in TRIOS software); optical microscopy; factorial and response-surface experimental design; data analysis, regression, and model validation.

Polymer and Composite Processing:

Compression molding, injection molding, extrusion, thermoforming, vacuum bagging, autoclave, out-of-autoclave and alternative cure methods, VARTM, RTM-light.